

Property-based Testing

Eine Einführung in FsCheck und CsCheck

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Patrick Drechsler

- Software Developer
- Work: C#
- Interests:
 - Software Crafting
 - Test-Driven Development
 - Functional Programming
 - Domain-Driven Design
- Slides are online: See QR-Code





Roadmap

- Problem w/ Example-Based Testing
- Examples solving the problem manually
- Property-Based Testing Frameworks
 - Vocabulary w/ examples
 - introduction to Generators & Shrinkers
 - more examples w/ Generators
- PBT Strategies
- Summary



Example based tests

Example:

```
public int Add(int a, int b)
    => a + b;
```

C#

Unit test:

```
[Fact]
public void Add_works()
    => Add(1, 2).Should().Be(3);
```

C#

More examples with parameterized test:

```
[Theory]
[InlineData(1, 2, 3)]
[InlineData(0, 1, 1)]
public void Add_works(int a, int b, int expected)
    => Add(a, b).Should().Be(expected);
```

C#

DEVELOPER FROM HELL



👹 Developer From Hell (DFH)

- Test-Driven Development (TDD)
- Only implement enough to make the test pass

DEVELOPER FROM HELL



👹 DFH 1/3

```
[Fact]
public void List_123_returns_321()
{
    List<int> input = [1, 2, 3];
    var actual = MyReverse(input);
    List<int> expected = [3, 2, 1];
    Assert.Equal(expected, actual);
}
```

```
// 👹
public List<int> MyReverse(List<int> input)
    => [3, 2, 1];
```


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👹 DFH 2/3

```
[Fact]
public void Empty_list_returns_empty_list()
{
    List<int> input = [];
    var actual = MyReverse(input);
    Assert.Equal([], actual);
}
```

```
// 👹
public List<int> MyReverse(List<int> input)
{
    if (input.Count == 0)
        return [];

    return [3, 2, 1];
}
```

DEVELOPER FROM HELL



👹 DFH 3/3

```
[Fact]
public void List_678_returns_876()
{
    List<int> input = [6, 7, 8];
    var actual = MyReverse(input);
    List<int> expected = [8, 7, 6];
    Assert.Equal(expected, actual);
}
```

```
// 👹
public List<int> MyReverse(List<int> input)
{
    if (input.Count == 0)
        return [];
    if (input.SequenceEqual((List<int>)[6, 7, 8]))
        return [8, 7, 6];

    return [3, 2, 1];
}
```



Let's find some properties for reversing a list

- result list has the same size
- result first element is last element of input list
- an empty list returns an empty list
- reversing a list twice returns the original list
- etc...



Live Coding

- Setup **without** a framework:
 - Adding 2 numbers
 - Reversing a list





PBT Framework Basics

■ Generator

- describe the input data!

■ Shrinker

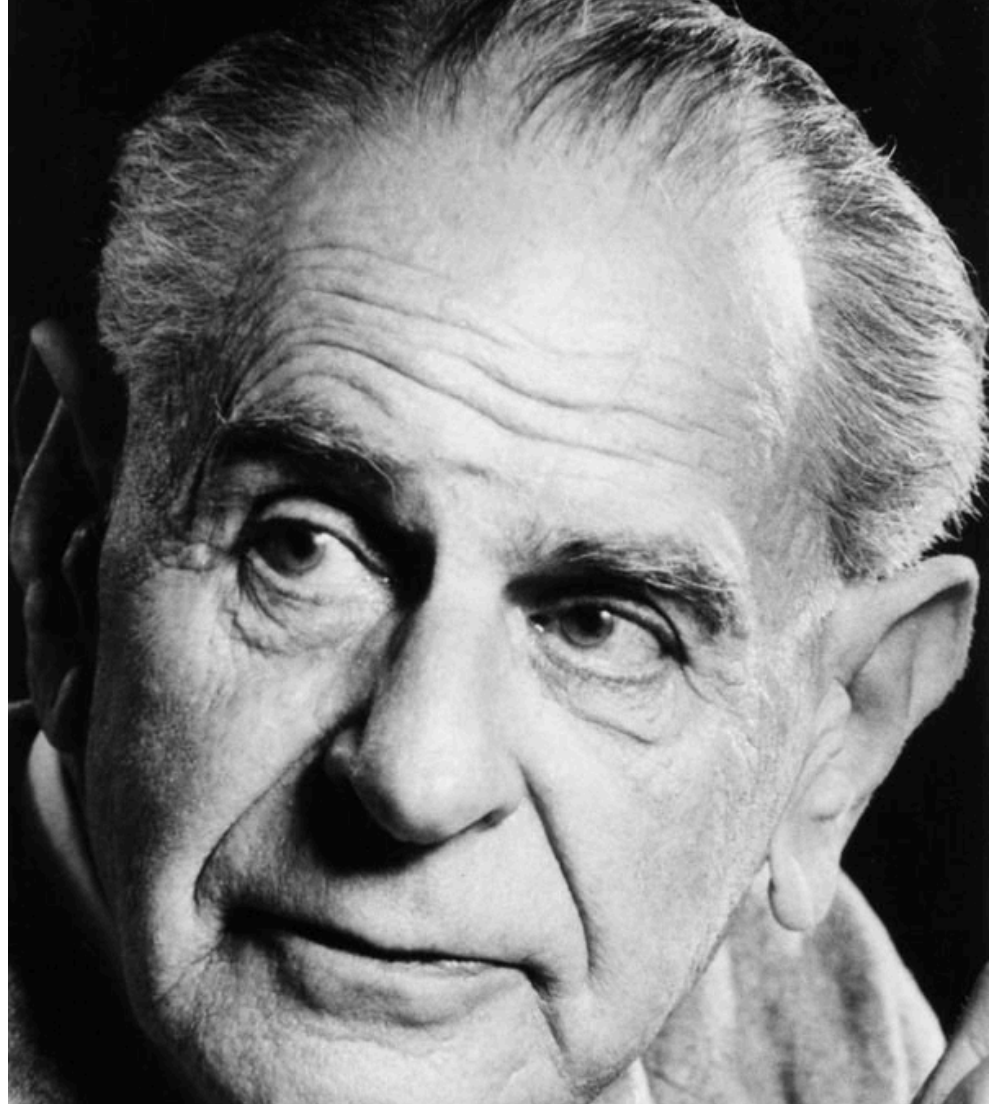
- framework gives us minimal examples

■ Generator + Shrinker 🙌 Arbitrary

- If something fails we not only get a falsifiable result: We get the closest result that does not fail.
- Some PBT Frameworks are called "Hypothesis" (i. e. in Python). Why?

Philosophy and natural sciences teach us...

- we can't prove anything
- but we can **falsify** a **Hypothesis**
- finding a good hypothesis is the difficult part







Hello World FsCheck (1/2)

```
using FsCheck;          // ⚠️
using FsCheck.Fluent;   // ⚠️

[Fact]
public void Reversing_a_list_twice_gives_the_original_list_v1()
{
    // This "Property" must return bool
    static bool CheckFn(List<int> list)
    {
        return MyReverse(MyReverse(list)).SequenceEqual(list);
    }

    // The lambda creates the test data input `list`
    Prop.ForAll((List<int> list) => CheckFn(list)) // ⚠️
        .QuickCheckThrowOnFailure();             // ⚠️
}
```

C#

- `Prop.ForAll`` : creates sample data
- `QuickCheckThrowOnFailure`` : throws on first failure



Hello World FsCheck (2/2)

```
using FsCheck;          // ⚠️
using FsCheck.Fluent;   // ⚠️
using FsCheck.Xunit;    // ⚠️

[Property] // ⚠️
public bool Reversing_a_list_twice_gives_the_original_list_v2(List<int> list) // ⚠️
{
    var actual = MyReverse(MyReverse(list));
    var expected = list;
    return actual.SequenceEqual(expected);
}
```

C#

- use ``Property`` attribute
- test must return ``bool``
- test must have input parameter(s)



Generators

- data is generated via reflection by default
- generators can be fine-tuned: <https://fscheck.github.io/FsCheck//TestData.html>
- Example: `Gen.Choose``

```
// Generate random numbers between 1 and 100
// which are divisible by 3
var arb = Arb.From(
    Gen.Choose(1, 100)
        .Select(x => x * 3));
```

C#

- don't forget to wrap the Generator in an Arbitrary...



Shrinking

```
private static bool IsValid(string s)
{
    return s.StartsWith("a");
}

[Property]
public void Validation_works(string input)
{
    Assert.True(IsValid(input));
}
```

C#

■ returns a **minimal** falsifiable example:

```
Falsifiable, after 1 test (2 shrinks) (15234498303011185687,2773856807235509173)
Last step was invoked with size of 2 and seed of (16350581963149029034,15962072172286382299):
Original:
"\012" (At least one control character has been escaped as a char code, e.g. \023)
Shrunk:
""
```



Live Coding

■ FizzBuzz (C#, F#)





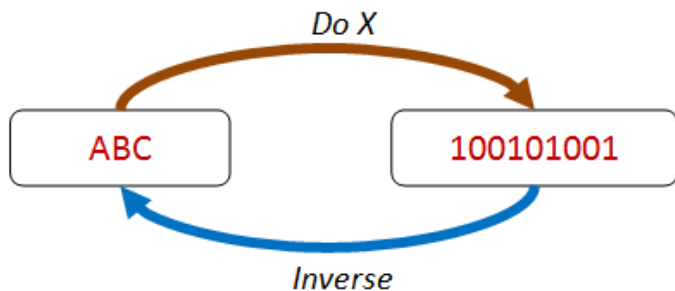
Strategies: "There and Back Again"

- When you have a forward and reverse function, i. e. Serialize/Deserialize
- Example: Reversing a list twice returns the original list

[<Property>]

```
let ``reversing a list twice returns the original list`` (aList: int list) =  
  let actual = aList ▷ List.rev ▷ List.rev  
  let expected = aList  
  actual = expected
```

F#



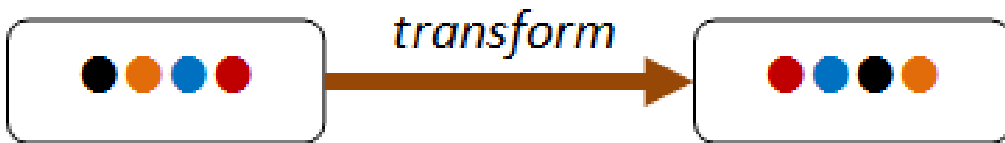


Strategies: "Some Things Never Change"

- When there is an invariant available
- Example: Sorting or Mapping a list never changes the length of the list

```
[<Property>]  
let ``sorting a list does not change its size`` (aList: int list) =  
  let actual = aList ▷ List.sort ▷ List.length  
  let expected = aList.Length  
  actual = expected
```

F#



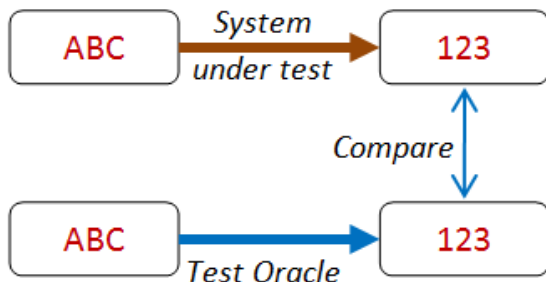


Strategies: "Test Oracle"

- When 2 functions should return the same result
- Example: Compare output of ``LegacyFn`` with ``RefactoredFn``

```
[<Property>]
let ``test oracle example`` (c: int, d: int) =
  let add1 a b = a + b
  let add2 a b = a * b
  let actual1 = add1 c d
  let actual2 = add2 c d
  actual1 = actual2
```

F#



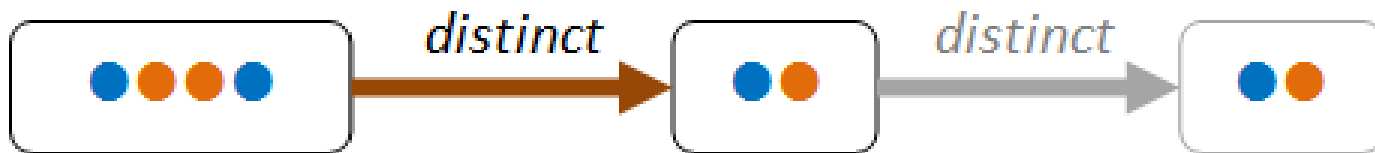


Strategies: "The More Things Change, The More They Stay The Same"

- When there are idempotent properties
- Example: Adding `0`, multiplying by `1`

```
[<Property>]
let ``adding zero is does not change the input`` (number: int) =
  let add a b = a + b
  let actual = add number 0
  let expected = number
  actual = expected
```

F#





Strategies: "Nuclear Exception"

- Just verify that the function does not throw an exception
- Example: Generate valid inputs, and assert that no exception is thrown

```
[<Property>]  
let ``nuclear explosion - function does not throw`` (str: NonEmptyString) =  
    let fn s = if String.IsNullOrEmpty(s) then failwith "ups" else s  
    fn str.Get = str.Get
```

F#



Generators: Also for custom types

- Since most PBT frameworks use reflection under the hood, any type can be generated.
- Example: Gilded Rose

```
type ItemArb =  
    static member Generate() =  
        gen {  
            let! name =  
                Gen.elements [  
                    "Aged Brie"  
                    "Sulfuras, Hand of Ragnaros"  
                    "Backstage passes to a TAFKAL80ETC concert"  
                ]  
            let! sellIn = Gen.choose (1, 100)  
            let! quality =  
                Gen.choose (1, System.Int32.MaxValue)  
            > Gen.map PositiveInt  
            return generateItem name sellIn quality  
        }  
    > Arb.fromGen
```

F#



PBT Frameworks for .NET

- FsCheck: The most popular
- Hedgehog: also worth a look
- CsCheck: the new kid on the block

All can be used from C# and F#.

The frameworks mainly differ in syntax.



Available for most languages

Just search for "Property Based Testing" and your language!

- Haskell: QuickCheck
- Python: Hypothesis
- JavaScript: fast-check
- Java: jqwik
- Scala: ScalaCheck
- Clojure: Test.check
- etc



When shouldn't you use PBT?

- When you don't have standard unit tests
- When standard unit tests are sufficient
- When you want to test a specific case
- When you can't identify suitable properties
- When your models are not suitable
- When your SUT is slow



Summary

- Property-based testing is a technique for testing statements of the type: **For all x that satisfy some precondition then some predicates holds**
- It can give you confidence that your code behaves correctly across a wider range of inputs.
- It can help you find bugs in your code resulting from inputs you never would have thought to test.
- This technique can be used alongside existing unit tests.



Thank You!

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Slides 

- QR Code or
- <https://draptik.github.io/2025-05-swec-property-based-testing/>
- sample code: <https://github.com/draptik/2025-property-based-testing>

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